



Jeril Kuriakose

Principal Data Scientist — Generative AI / LLMs

Deep expertise in LLMs, NLP and agentic AI systems. I lead end-to-end development of high-impact platforms — from data engineering and training through MLOps, inference optimisation and secure deployment at scale.

Currently at SDAIA, owning the post-training and inference stack for ALLaM, Saudi Arabia's national Arabic LLM. Open to principal and staff-level roles in applied LLM research.

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14

YEARS IN DATA & AI

100K

PERSON-HOURS SAVED / YR

35+

PEER-REVIEWED PAPERS ↗

01 Practice

Current focus and working stack

Recent focus is **Arabic LLMs (ALLaM)**, agent orchestration, and the post-training stack: curriculum SFT, iterative DPO, and RL-SFT hybrid training with gradient separation across objectives.

Alongside it, large-scale inference on Kubernetes with vLLM, TGI and Triton, plus Ray and Azure. Proven record of measurable business outcomes and hands-on leadership of cross-functional data teams.

ML / AI

LLMsNLPAgentic systems

RAGTime-series

LANGUAGES & FRAMEWORKS

PythonPyTorchTransformers

TRLData-JuicervLLMRay

LangGraphDify

MLOPS, INFRA & DATA

KubernetesAzureMLflow

ClickHouseQdrant

AI - AUGMENTED DEV

Claude CodeOpenCodeCodex

Custom skills & agents

2024

Now

Principal Data Scientist Saudi Data & AI Authority **CURRENT**

SDAIA · Riyadh, Saudi Arabia Jan 2024 – Present

Owns the **post-training and inference stack for ALLaM**, Saudi Arabia's national Arabic large language model, from **50TB**-scale data processing through 70B-parameter production serving.

- Lead data processing and inference optimisation for **ALLaM** (Arabic LLM); manage training and inference stacks across PyTorch, Transformers, TRL, vLLM/TGI/Triton, Ray, Kubernetes and Azure.
- Built the post-training stack for agentic capability: interleaved RL with periodic SFT injections, iterative DPO with uncertainty-aware preference collection, self-optimised fine-tuning via self-distillation, and gradient separation across RL/SFT objectives.
- Designed a curriculum SFT pipeline for agentic behaviour; improved model convergence by **~35%**.
- Processed **50TB** of RedPajama-Data-v2 through Data-Juicer; implemented LLaMA-2 pretraining strategies, both from scratch and continual.
- Deployed and optimised ALLaM **7B / 13B / 70B** on Kubernetes; identified optimal inference parameters per use case.
- Implemented agentic RAG and multi-agent planning and orchestration with LangChain, LangGraph and Dify.
- Designed AI-augmented development workflows on Claude Code, OpenCode and Codex; authored custom skills and agents that automate engineering tasks and shorten delivery.

2019

2023

Senior Data Scientist Mizuho Bank

Singapore Jul 2019 – Dec 2023

Built and productionised **four named NLP and forecasting platforms** for a global investment bank, and introduced the organisation's first model-tracking practice.

- **Phoenix** — generic NLP information-extraction platform on BERT with a hybrid CNN and Bloom embeddings; **~90%** accuracy; saved **~100,000 person-hours per year**.
- **PIGEON** — document classification with RoBERTa plus summarisation with GPT-2 and GPT-Neo; **97%** accuracy; full API productionisation.
- **SWAN** — anti-money-laundering name screening; cut false positives by **20%**; vector database for embeddings and graph neural networks for anomaly detection.
- **HAWK** — hybrid encoder-decoder LSTM for daily, weekly and monthly forecasts; saved **~3,000 person-hours per year**; external data ingestion with ELK.
- Introduced **MLflow** for organisation-wide model tracking; implemented ML monitoring and production performance validation.

2019
6 mos

Senior Data Scientist **Baker Hughes**

a GE Company · Kochi, India Jan – Jun 2019

- Predicted **stuck-pipe events** with feature-engineered ensemble models; identified the key drivers and the timing of incidents.
- Analysed non-productive time and drill-bit wear; delivered actionable insight to reduce rig downtime.
- Drove awareness of applied data science among subject-matter experts through targeted training.

2018
2019

Senior AI Engineer **Innovation Incubator**

Thiruvananthapuram, India Feb 2018 – Jan 2019

- Generic keyword extraction across PDF, TXT, Excel and HTML via ensemble ML and feature engineering; shipped production APIs.
- Customer segmentation and claim-settlement prediction; improved healthcare revenue-cycle management by **~18%**.
- Owned end-to-end ML pipelines and backend services; worked directly with customers to translate business problems into ML problems.

2017
2018

Senior ML / AI Developer **Raw Data Technologies**

Kochi, India Apr 2017 – Feb 2018

- EMI defaulter prediction with ensemble models and extensive feature engineering; deployed APIs and data pipelines.
- Performance analysis of internal versus external collection agents using the Kaplan-Meier estimator and cumulative incidence.

2012
2017

EARLIER – ACADEMIA & ENGINEERING

2015-2017 **Assistant Professor & Data Analyst** · St. John College of Engineering and Technology, Palghar

2013-2015 **PhD Scholar, Developer & Data Analyst** · Manipal University Jaipur

2012-2013 **Developer & Assistant Professor** · Kavary College of Engineering, Salem

03 Selected publications

From 35+ total · see Google Scholar

2025

ALLaM: Large Language Models for Arabic and English

Bari M. S., Kuriakose J., et al. ICLR

2023

A review of deep learning-based approaches for detection and diagnosis of diverse classes of drugs

Kuriakose J., et al. Archives of Computational Methods in Engineering (Springer)

2022

EMBN-MANET: Eliminating Malicious Beacon Nodes in UWB-based Mobile Ad-Hoc Networks

Kuriakose J., Joshi S., Bairwa A. K. Ad Hoc Networks (Elsevier)

2015

Secure Multipoint Relay Node Selection in Mobile Ad Hoc Networks

Kuriakose J., Amruth V., Raju R. V. Communications in Computer and Information Science (Springer)

2014

A Review on Mobile Sensor Localization

Kuriakose J., et al. Security in Computing and Communications (Springer)

04 Credentials

Education, recognition and languages

EDUCATION

Ph.D. in Computer Engineering

Manipal University Jaipur, School of Computing and IT

Jul 2013 – Dec 2019

M.Tech. in Information Technology

ISIM, University of Mysore

Jul 2012

B.Tech. in Information Technology

Jeppiaar Engineering College (Anna University)

May 2010

LANGUAGES

Tamil	NATIVE
Malayalam	NATIVE
English	FLUENT
Hindi	FLUENT
Arabic	BASIC

RECOGNITION

🏆

Project of the Year

Mizuho Bank · Jul 2022

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Honourable Award – AML Project

Mizuho Bank · Feb 2020

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Employee of the Month

Mizuho Bank · Oct 2019, May 2021, Jul 2021

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Honourable Mention – Phoenix Project

Mizuho Bank · Dec 2020